The Sensitivity Peak of a PI Loop from a Single Step Response

Abstract

The sensitivity peak M_s measures how robust a control loop is. It is defined in the frequency domain, so measuring it needs a plant model or a frequency sweep, and most installed loops have neither. This paper obtains M_s from a single ordinary closed-loop setpoint step, with no model and no dedicated experiment. The error record of a setpoint step is the step response of the sensitivity function, because $E(s) = S(s)/s$. Counting the turns N that the error trajectory makes about its settling point gives the damping of the dominant mode. One further number, the slope ν of the loop at its $-180°$ crossing in the coordinate $w = -\ln(-L)$, converts that damping into the peak. The result is $M_s|\nu| = N/\kappa$, from which the distance to instability cancels. When the integral time cancels the dominant lag, $\nu = 1+j\pi/2$ exactly, whatever the dead time, so no plant knowledge is needed. A record winding less than about three quarters of a turn then lies inside the classical window $M_s \in [1.2, 2.0]$. The reading is valid whenever the error changes sign, a condition read from the same record.

1 Introduction

The sensitivity peak $M_s = \max_\omega |S(j\omega)|$, with $S = 1/(1+L)$ and loop transfer function $L = CG$ formed from the controller C and the plant G, says how robust a loop is. A value near 1.4 is comfortable, a value near 2.0 is marginal [2, 3, 5].

Measuring it is the problem. M_s is defined in the frequency domain, so it needs the loop transfer function. That means a model somebody must identify, or a sweep somebody must run. The installed base has neither. Of more than eleven thousand controllers surveyed by Desborough and Miller [4], 97% use a PID algorithm, and by their account most do not perform well.

What plants do monitor from routine data is something else. Hägglund's monitor finds oscillation [7] and his idle index finds sluggishness [8]. The Harris index and its successors compare variance against minimum-variance control [9, 10]. A survey of industrial practice [11] confirms the gap: performance is watched, robustness is not. The quantity chosen at design time is the one nobody checks again.

This paper measures M_s from a single ordinary setpoint step. No model is identified, no relay test is run, no frequency is swept, and the loop stays in service. The result is one relation. Let N count the turns the error makes about its settling point, in the lowest of two frequency bands that shows ringing, and let ν be the slope of the loop where its phase passes $-180°$. Then

$$M_s\,|\nu| = \frac{N}{\kappa}, \tag{1}$$

where κ comes from the counting rule. The distance to instability cancels from the derivation, which is why a record is enough. For a PI controller whose integral time cancels the dominant lag, ν is fixed at $1+j\pi/2$ whatever the dead time, and nothing about the plant is needed.

Each part of (1) has neighbours; the combination does not. Relay autotuning [6] and M_s-based design [2] reach the same information, but through a dedicated experiment or an identified model. Model-free routine-data indices [7, 8] need neither, but report oscillation or sluggishness instead of a peak. Minimum-variance monitoring [9, 10] uses routine data, but measures variance and needs a dead-time estimate. Recent data-driven work bounds gain and phase margins without a model [12], though from designed data through an optimisation. Data-driven tuning [13, 14, 15, 16] adjusts controllers rather than measuring them.

Closest in spirit are the classical readings of a transient: overshoot, decay ratio, and the logarithmic decrement. All three fit a damping ratio to two peaks of an assumed second-order response. The turn count uses the shape of the whole trajectory, and it gives M_s rather than a damping ratio.

The controller is PI. Derivative action is treated in Section 6, where it is shown to cost three separate assumptions while adding nothing the method needs. The plant is stable, self-regulating, with monotone gain and phase. The counting device is the turn index of [1], used there as a damping diagnostic. So that the three kinds of statement do not mix, Section 2 is exact and holds no measured quantity, Section 3 names the approximations, and Section 4 holds every number.

2 Theory

This section is exact. Every claim is proved and no measured quantity appears. The argument runs in three moves: the record is the sensitivity function, counting turns measures the damping of its dominant pole, and one complex number turns that damping into the peak.

Proposition 1. *For a unit setpoint step on a one-degree-of-freedom loop, $E(s) = S(s)/s$.*

Proof. $E = R - Y = R(1 - T) = RS$ with $R = 1/s$. ☐

The error record is therefore the step response of S. One ordinary step is enough, and no second experiment is needed. This is also where the restriction to unit setpoint weighting enters.

Next, what the count measures. Write e for the control error, Δe for its difference between samples, and E for its running integral. Following [1], form two phase portraits and count the turns each makes about the settling point, with each axis scaled by its own peak and the curve cut at its last entry into a disc of radius ε [1, Def. 1]:

$$N_0 \text{ from } (e,\, E - E_{\text{end}}), \qquad N_1 \text{ from } (\Delta e,\, e). \tag{2}$$

Integrating emphasises slow content and differencing fast content, so N_0 answers to the lower part of the spectrum and N_1 to the middle. ([1] adds a third portrait from the second difference, which points to the derivative channel. A PI loop has none, and that count stays at zero throughout our tests.)

Each band has its own limit, $\bar{N}_0 = 0.5$ and $\bar{N}_1 = 0.75$ [1]. Write $k_{\min}$ for the lowest band whose count exceeds its limit. The reading is taken there:

$$N := N_{k_{\min}}. \tag{3}$$

Two reasons fix this choice, and Section 4 measures both. Integration suppresses fast modes, so a resonance well above the integral crossover can be invisible in N_0 while N_1 sees it plainly. And differencing amplifies noise, so where N_0 does suffice it is by far the steadier of the two. Scanning upward from the lowest band takes the steadier count whenever it is informative, and falls through to the next only when it is not. For a mode $e^{-\alpha t}\cos\omega t$, of decay rate α and ringing frequency ω,

$$N = \kappa\,\frac{\omega}{\alpha} + \eta, \qquad \kappa = \frac{\ln(1/\varepsilon)}{2\pi}, \tag{4}$$

which is [1, Prop. 3]. Here η is a small term from the two cut ends, bounded in Proposition 2. The default $\varepsilon = 0.1$ gives $\kappa = 0.3665$, and it is used throughout. The count is a pure number. It does not change if the step is larger or the loop faster (Figure 1).

Proposition 2 (Endpoint bound)**.** *For the step response of the mode, $|\eta| < 1/2$.*

Proof. [1, Prop. 3] allows a half turn at each cut end, giving $|\eta| < 1$. For a unit step the mode is $e(t) = e^{-\zeta\omega_n t}\big(\cos\omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}}\sin\omega_d t\big)$, with damping ratio ζ, natural frequency ω_n and damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$, so $e(0) = 1$ and $\dot{e}(0) = 0$. The trajectory starts on the e axis for every ζ, so only the far end is free. ☐

So the count gives ω/α, the ringing of the dominant pole. Turning that into M_s takes one more step, and it is the heart of the paper.

Stability is lost through a single pole pair at the frequency ω_{180} where the phase passes $-\pi$. This is Proposition 2 of [1], taken unchanged. What follows is new.

Proposition 3 (Normal form)**.** *Let $w := -\ln(-L)$, on the branch continuous through ω_{180}, where $-L$ is positive real. Then*

$$1 + L = 1 - e^{-w} \tag{5}$$

for every loop and every frequency.

Proof. $e^{-w} = e^{\ln(-L)} = -L$. ☐

Everything below is read off (5). Two numbers describe w near the crossing: its value there, and how fast it changes,

$$\mu := w(\omega_{180}), \qquad \nu := \left.\frac{dw}{d\ln\omega}\right|_{\omega_{180}}. \tag{6}$$

The first is real, because $\arg L = -\pi$ there, and equals the log gain margin. The second is complex. Its real part is the magnitude slope and its imaginary part the phase slope, the pair that Bode's gain–phase relation ties together.

Proposition 4 (Pole)**.** *The closed-loop poles are where $w = 0$. If w is affine in $\ln\omega$, that is $w = \mu + \nu\ln(\omega/\omega_{180})$, then*

$$p = j\,\omega_{180}\,e^{-\mu/\nu}, \tag{7}$$

and $\omega/\alpha = \cot\beta$ with $\beta = \mu\,\mathrm{Im}\,\nu/|\nu|^2$. Both are exact.

Proof. By (5), $1 + L = 0$ iff $w = 0$. The affine form then gives $\ln(s/j\omega_{180}) = -\mu/\nu$, hence (7). Splitting $-\mu/\nu = -\mu\bar{\nu}/|\nu|^2$ into real and imaginary parts gives $p = \omega_{180}e^{-\mu\mathrm{Re}\nu/|\nu|^2}(j\cos\beta - \sin\beta)$. ☐

Proposition 5 (Peak)**.** $M_s^{-1} = \min_\omega |1 - e^{-w}|$. *Under the affine form, real frequencies carry w along the straight line through μ in the direction ν, and to first order in μ the minimum is the distance from the origin to that line,*

$$M_s = \frac{|\nu|}{\mu\,\mathrm{Im}\,\nu}. \tag{8}$$

Proof. The first claim restates the definition of M_s under (5). For the second, $|1 - e^{-w}| = |w| + O(|w|^2)$, and the origin lies $\mu|\mathrm{Im}\,\nu|/|\nu|$ from the line. ☐

The pole is where the line, continued off the real axis, reaches the origin. The peak is where the real line passes closest to it (Figure 2). Dividing one by the other removes μ, the distance to instability, which the record cannot report.

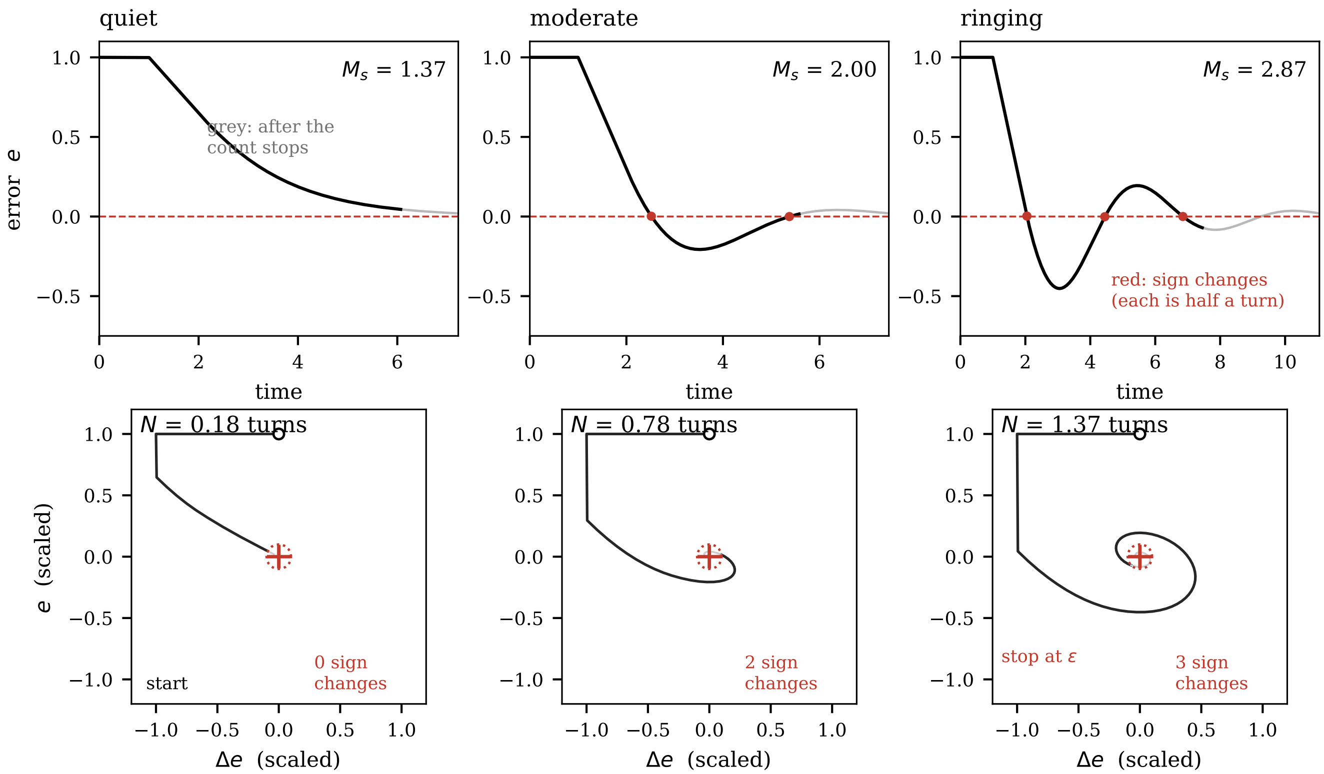

Figure 1: The counting rule (4) and the validity condition A3. Top: the error record, black where the count is taken and grey after it stops. Bottom: the same error against its own difference. The count is close to half the number of sign changes, so a record with none, as on the left, carries no reading.

Proposition 6 (The law)**.** *To first order in* μ*,*

$$M_s\,|\nu| = \frac{\omega}{\alpha} = \frac{N-\eta}{\kappa}. \tag{9}$$

Proof. $\cot\beta = 1/\beta + O(\beta)$, so $\omega/\alpha = |\nu|^2/(\mu\,\mathrm{Im}\,\nu) + O(\mu)$, which is $|\nu|$ times (8). The rest is (4). ☐

The count measures ω/α and nothing else. The plant enters only through $|\nu|$. The controller does not appear at all.

The law needs $|\nu|$, which the record does not give. Sometimes the loop fixes it.

A dead time L_d adds phase $-L_d\omega$, and its phase slope is $L_d\omega$. Its slope equals its phase. A lag behaves differently: its phase tends to $\pi/2$ while its slope falls back to zero.

Proposition 7 (Pinning)**.** *Let the plant be first-order plus dead time with lag* T *and delay* L_d*, and let the integral time satisfy* $T_i = T$*. Then* $\nu = 1 + j\pi/2$ *exactly, for every* L_d*, so* $|\nu| = \sqrt{1+\pi^2/4} = 1.862$*.*

Proof. With $T_i = T$ the controller zero cancels the plant pole, leaving $L = Ke^{-L_d s}/(Ts)$ with K the combined gain. Then $w = \ln(T\omega/K) + j(L_d\omega - \pi/2)$, so $\nu = 1 + jL_d\omega$. The crossing $\mathrm{Im}\,w = 0$ forces $L_d\omega_{180} = \pi/2$. ☐

A longer dead time only moves the crossing to a lower frequency (Figure 3). The product $L_d\omega_{180}$ does not move. Under Proposition 7 the law becomes $M_s = N/(\kappa|\nu|)$, with everything on the right known in advance.

3 What is approximate

Section 2 rests on one identity, (5), and three approximations. They are named here without numbers. Section 4 measures each.

A1. *w is affine.* Propositions 4 and 5 hold ν fixed over the band that matters. This is the only modelling step in the paper, since (5) is an identity. A real loop has ν varying, so the error should grow with $|d\nu/d\ln\omega|$, and with the span over which the affine form is used, which is set by μ.

A2. *First order in μ.* Propositions 5 and 6 linearise $|1-e^{-w}|$ and $\cot\beta$. Both are exact as $\mu \to 0$. A loop with a comfortable gain margin is not in that limit. Proposition 4 is free of this step, so its error measures A1 alone.

A3. *One dominant mode.* Equation (4) describes a single mode. When integral action is slow, a non-oscillating decay dominates the record instead, the

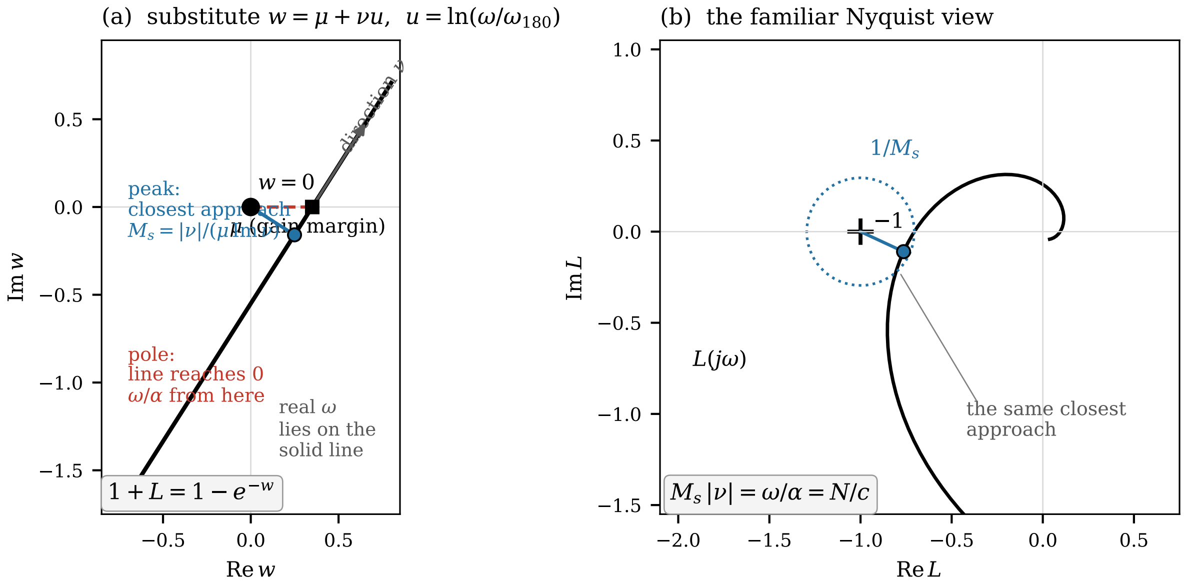

Figure 2: The construction in the coordinate $w = -\ln(-L)$. Real frequencies lie on a straight line. The pole is where it reaches the origin, the peak where it passes closest, and the ratio of the two is $|\nu|$.

resonant mode rides on it as ripple that never crosses zero, and the trajectory cannot wind. This failure is visible: *the reading is valid when the error changes sign*. The test needs no threshold and no model.

Three limits of scope are not approximations, and they are the paper's real exposure. First, the law needs $|\nu|$, and Proposition 7 supplies it only under its own hypotheses. Outside them $|\nu|$ must be assumed, and unlike A3 the record offers no test of whether the assumption is sound. Second, Proposition 1 assumes unit setpoint weighting. Two-degree-of-freedom controllers are common, and although the closed-loop poles do not move, so the law survives, the bound of Proposition 2 rests on $\dot{e}(0) = 0$ and does not. Third, the analysis is noise-free, while noise enters the differenced coordinates directly.

A fourth point concerns sampling, and it is standard sampled-data practice rather than a limit of the method. A sampled record reports the sampled loop, not the continuous one, and near the boundary the small extra phase from the hold matters in relative terms. The requirement is quantified in Section 4.

4 Evidence

The claims of Section 2 are of two kinds, and they are tested separately so that the evidence for each is as clean as the claim allows.

Propositions 3–6 relate the loop to its poles and to its peak. Nothing in them refers to a record, so they are tested by exact computation: closed-loop poles by root finding, M_s and ν from the frequency response. No simulation, no sampling, and no counting enters. Equation (4) and Proposition 2, by contrast, concern reading ω/α off a record, and are tested on an analytic mode whose value is known in closed form. Only the closing demonstration simulates a loop.

4.1 The law, computed exactly

Six plant families are used: first- and second-order plus dead time, a three-lag chain with dead time, and three delay-free chains of four and six equal lags and four unequal ones. Gain is swept from a margin of 1.6 to 1.01, giving 60 points with pole residuals at machine precision. Proposition 6 predicts $R := M_s|\nu|\,\alpha/\omega \to 1$.

μ	0.3–1.0	0.1–0.3	0.03–0.1	< 0.03
mean R	1.224	1.090	1.036	1.009
sd	0.058	0.022	0.009	0.005

R converges to one and its scatter falls with it. All six families lie on the same curve; the delay-free members are not distinguishable from the rest, which is what (9) requires, since the plant enters only through $|\nu|$.

Figure 4(a) tests the order. On logarithmic axes the measured slope is 1.054, against 1 for a first-order term and 2 for a second-order one, and a least-squares fit gives $R - 1 = 0.564\mu - 0.006$ with the intercept zero to within its scatter. The excess of R over one and the shortfall away from the boundary are one effect, so

$$M_s|\nu| = \frac{\omega}{\alpha}\big(1 + k\mu\big). \tag{10}$$

The size of k is predicted too. Approximation A1 freezes ν, so its error should grow with how fast ν really

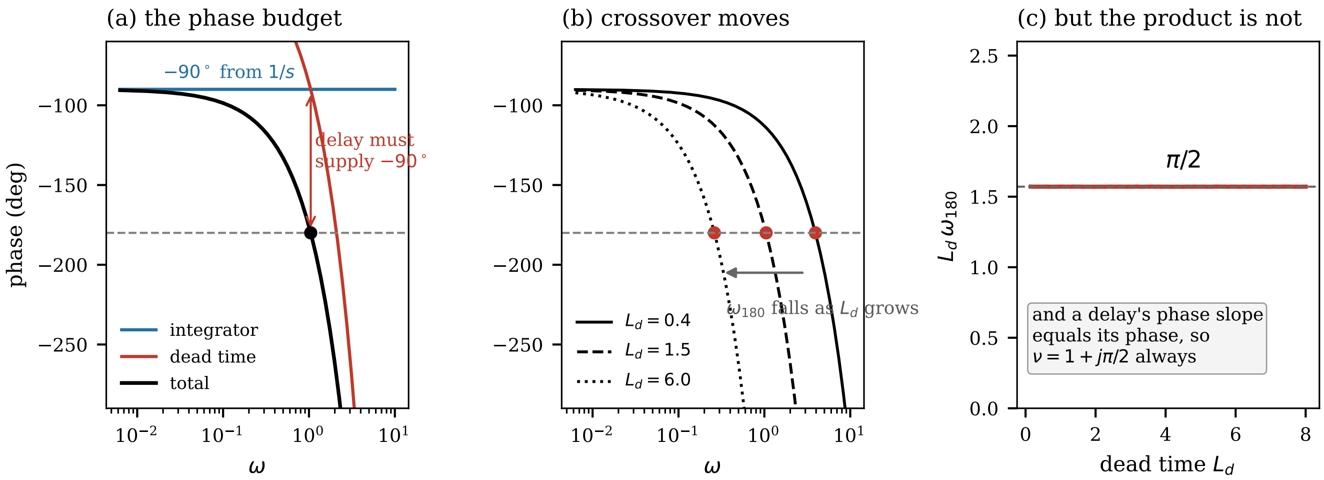

Figure 3: Proposition 7. The integrator takes half the phase budget, so the delay must take the other half, whatever its size. This fixes ν.

varies. Figure 4(b) plots the fitted k against $|d\nu/d\ln\omega|$ for the six families; the correlation is $r = +0.83$.

4.2 The counting rule

On the analytic step response of a single mode, over $\zeta \in [0.05, 0.75]$ at 120 points, the count satisfies $\max|\eta| = 0.23$ against the bound of 1/2 proved in Proposition 2. The three bands agree with one another to within that term, as [1, Prop. 3] requires: mean offsets are -0.065 between bands 0 and 1 and $+0.072$ between bands 1 and 2.

Sampling is the one practical caveat, and it is a sampled-data matter rather than a limit of the method. The count of a sampled record reports the sampled loop, which the hold and the quantised dead time make slightly different from the continuous one; near the boundary, where the damping is small, that difference is large in relative terms. The requirement is therefore stated per cycle of the ringing and grows with the number of turns: at 1500 samples per cycle a loop winding 25 times still reads a few percent low, and the shortfall halves as the step is quartered.

4.3 Why the reading is taken at $k_{\min}$

Two measurements fix the rule of (3). First, N_0 can miss a genuine resonance. On a lag-dominant plant at $\tau = 0.05$ with $T_i/T = 0.3$, where the crossing sits at $\omega_{180}T_i = 8.5$, a loop with $M_s = 2.58$ gives $N_0 = 0.36$, below its limit, while $N_1 = 0.83$ is above its own. Second, N_1 is fragile under noise where N_0 is not. At 300 samples per transient on a loop with $M_s = 2.13$, one percent of noise leaves N_0 at 0.73 ± 0.01 but drives N_1 from 0.83 to 0.21 ± 0.20: differencing lets noise dominate the Δe axis, the trajectory enters the truncation disc at once, and the winding is lost. Scanning upward from band 0 takes the robust count when it is informative and the sensitive one when it is not.

4.4 End to end

The remaining tests simulate a loop, and are demonstrations rather than evidence for the law. Each lag is integrated exactly under zero-order hold. On the validity condition, over 497 runs on four process-typical plants, records whose error never changes sign never gave a count above 0.32 of the limits of [1], while the true M_s of those loops ranged from 1.2 to 6.0; the reading is therefore one-sided. A load step does not recover the lost mode, giving $N = (0.13, 0, 0)$.

On accuracy: four plants [1], $T_i/L_d \in \{1.5, 2, 3, 4, 6\}$, four gain levels, one setpoint step each, against exact M_s. On valid records the median error is +5%, the interquartile range −7% to +8%, and 22 of 24 readings fall within ±25%. In a matched test over five classical tuning rules, every gross error (−51%, −47%, −37%, −31%) fell on a record the validity condition rejects, while accepted readings stayed within −10% to +14%.

5 Use in practice

When the integral time cancels the dominant lag, the method is three steps. Step the setpoint and record the error until it settles. Check that the error changes sign; if it does not, the record gives no reading. Count the turns in the lowest band that exceeds its limit.

Table 1 converts the count. The classical window $M_s \in [1.2, 2.0]$ is a count between 0.15 and 0.78 in band 1, and between 0.04 and 0.66 in band 0, which gives a rule an engineer can apply by eye:

Under about three quarters of a turn, the loop is

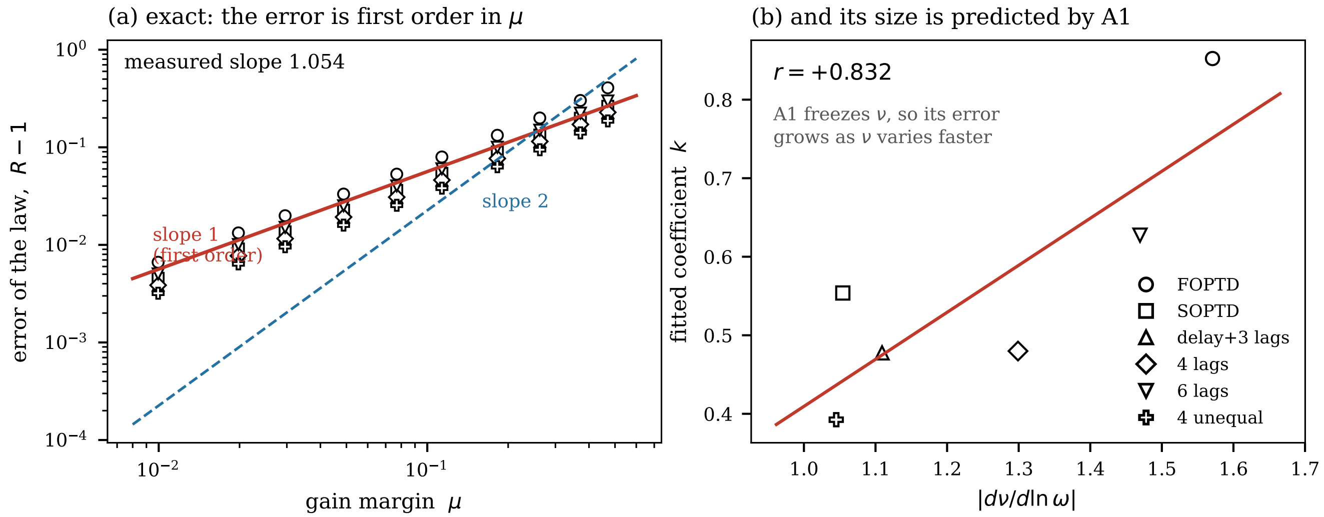

Figure 4: Two tests of the error term, from exact computation. (a) The measured slope is 1.05, so the error is first order in μ. (b) Its coefficient is set by how fast ν varies, as A1 predicts.

Table 1: Reading the count, from the calibration of Section 4, with the dominant lag cancelled. The band matters little, but it should be recorded with the count.

	M_s if read in		
turns	band 0	band 1	
0.2	1.6	1.4	comfortable
0.4	1.7	1.7	acceptable
0.8	2.2	2.0	marginal
1.5	3.1	3.0	too hot

inside the classical robustness window. Beyond that, it is not.

Three uses follow. Loops receive setpoint changes as a matter of routine, so each one becomes a robustness sample, and a loop drifting towards the stability boundary shows a rising count before it causes an incident. A relay experiment [6] answers the same question but must force sustained oscillation, which is often unacceptable on a live unit; one setpoint step avoids it. And because the count is a pure number, counts from a flow loop and a temperature loop compare directly, so a plant's loops can be ranked without tuning any of them.

Are the three limits of [1] reasonable? The tuning rule of [1] uses the same bands and the same $k_{\min}$, but to a different end: it corrects the gain of that band, whereas we read the peak there. Its limits $\bar{N} = (0.5, 0.75, 1.0)$ were fixed by calibration, and the law lets them be examined.

One might expect the three limits to differ because the three portraits count differently. They do not. Proposition 3 of [1] shows that all three portraits are linear images of one spiral, and measurement agrees. For a pure dominant mode the counts are equal to within the endpoint term, the mean offsets being -0.065 and $+0.072$ against ladder steps of 0.250.

Table 2: Three counts for one pure mode, and the bands the ladder calls violated. The counts agree; the limits do not.

ζ	N_0	N_1	N_2	violated
0.60	0.53	0.44	0.49	$\{0\}$
0.45	0.80	0.75	0.80	$\{0, 1\}$
0.30	1.39	1.33	1.36	$\{0, 1, 2\}$

Table 2 shows what the ladder buys. Because the counts agree while the limits rise, the lowest band is always violated first, and the violated set fills in from below. This is what makes $k_{\min}$ carry information. Were the limits equal, one mode would cross all three at the same instant, every band would be flagged together, and the attribution step of the tuning rule would have nothing to attribute.

The limits are also sensible as a specification. For a loop with one dominant mode the binding limit is the lowest, $\bar{N}_0 = 0.5$, which is a peak of 1.71: inside the classical window. The higher limits bind only when a faster mode is present that the lower bands cannot see, and there the admissible peak rises to 1.92 and 2.29. Tolerating more at high frequency is the right choice, because by Proposition 4 of [1] a low-frequency mode leaks upward into every band above it. An upper count is the more likely to be inflated, so it needs the looser guard.

6 Derivative action

The law itself does not care about the controller. The derivation uses only the loop L and its slope at one frequency, and the experiment above confirms it: with T_d/T_i at 0, 0.15 and 0.30, the ratio R is 1.183, 1.204 and 1.213.

What derivative action costs is everything around the law. Three assumptions are added, and none is repaid.

First, Proposition 7 is lost. Its proof needs the integrator to supply $-\pi/2$ and the delay to supply the rest. Derivative action adds phase lead, so the delay supplies less, by an amount that depends on how much delay there is. The slope stops being fixed:

T_d/T_i	$\|\nu\|$ at $\tau =$			spread
	0.3	0.5	0.9	
0 (PI)	1.862	1.862	1.862	0%
0.15	1.878	1.754	1.865	7%
0.25	2.290	1.645	1.868	34%
0.40	2.676	1.871	1.871	42%

So $|\nu|$ must be supplied from outside, and the reading is no longer model-free.

Second, a free constant appears. Derivative action needs a filter, and $|\nu|$ depends on its pole: filter ratios of 5, 10 and 20 give $|\nu| = 1.655$, 1.645 and 1.643 at $T_d/T_i = 0.25$. The differences are small, but the constant must be declared, and it is not one the record reveals.

Third, Proposition 2 loses its proof. The bound $|\eta| < 1/2$ follows from $\dot{e}(0) = 0$. With derivative action on the error, a setpoint step puts an impulse into the control, and the trajectory no longer starts on the axis. The weaker bound $|\eta| < 1$ of [1] is all that remains, unless the derivative is taken on the measurement, which changes the record being read.

None of this bears on the law. It bears on what must be known before the law can be used, and with derivative action that is a supplied $|\nu|$, a declared filter constant, and a weaker endpoint bound. For PI, all three are free.

7 Conclusion

A setpoint step contains the sensitivity function, because $E(s) = S(s)/s$. In the coordinate $w = -\ln(-L)$ the loop becomes $1 + L = 1 - e^{-w}$, an identity. When ν is steady, real frequencies carry w along a straight line: the pole is where that line reaches the origin, the peak where it passes closest, and the ratio of the two is $|\nu|$. The distance to instability cancels, leaving $M_s|\nu| = N/\kappa$.

With the lag cancelled, $\nu = 1 + j\pi/2$ exactly, whatever the dead time, because a delay contributes equally to phase and to phase slope. The reading then needs nothing about the plant. What is left for the engineer is short: step the setpoint, check the error changes sign, count the turns. Under about three quarters of a turn the loop is inside the classical robustness window. A quantity chosen at design time, and never checked since, can be read from any routine setpoint change.

Two conditions bound the result, both stated rather than hidden. The analysis is noise-free, and noise enters the differenced coordinates directly. And the reading is one-sided, valid when the error changes sign and silent otherwise.

Data availability

Scripts reproducing every number are included with the SPIN reference implementation (`https://github.com/ottapav/roboPID-simulator`).

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